

# DATA CLEANING & PREPARATION PROJECT

## Internship Project Report

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**Tools Used:**

- Microsoft Excel

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### 1. Project Overview

The objective of this project was to clean and prepare an e-commerce dataset for analytical processing. The dataset contained customer order information including product details, payment methods, coupon codes, order status, and pricing information.

The project focused on improving data quality by identifying missing values, validating records, checking for duplicate entries, and ensuring formatting consistency across the dataset.

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### 2. Dataset Information

| Detail        | Information               |
|---------------|---------------------------|
| Dataset Type  | E-commerce Orders Dataset |
| Total Records | 1200                      |
| Total Columns | 14                        |
| Tool Used     | Microsoft Excel           |

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### 3. Understanding the Dataset

The dataset contains multiple business-related fields including:

- Order ID
- Customer ID
- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- Order Status

- Coupon Code
- Total Price

These fields are commonly used in business analytics and sales reporting.

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#### **4. Step 1: Missing Value Analysis**

The dataset was inspected to identify null or missing values.

**Observation:**

A total of 309 missing values were identified in the CouponCode column.

**Action Taken:**

To maintain consistency and avoid blank entries, all missing values were replaced with:  
"No Coupon Used"

**Result:**

The dataset became more complete and suitable for further analysis.

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#### **5. Step 2: Duplicate Record Verification**

The dataset was checked for duplicate records using Excel's built-in duplicate detection tools.

**Observation:**

No duplicate records were found in the dataset.

**Result:**

No rows needed to be removed.

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#### **6. Step 3: Data Format Validation**

The dataset was reviewed for formatting consistency.

**Validation Performed:**

- Verified date formatting
- Checked numerical values
- Reviewed text fields for consistency

**Result:**

The dataset maintained proper formatting standards suitable for analytical processing.

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### 7. Project Change Log

| Change ID | Description                                            | Status    |
|-----------|--------------------------------------------------------|-----------|
| C001      | Identified missing values in CouponCode column         | Completed |
| C002      | Replaced blank CouponCode values with "No Coupon Used" | Completed |
| C003      | Checked dataset for duplicate records                  | Completed |
| C004      | Verified date and numeric formatting                   | Completed |
| C005      | Prepared cleaned dataset for analysis                  | Completed |

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### 8. Learning Outcomes

Through this project, practical understanding was developed in:

- Data preprocessing
  - Missing value handling
  - Duplicate verification
  - Data validation
  - Excel-based cleaning workflows
  - Project documentation
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### 9. Conclusion

The dataset was successfully cleaned and validated for analytical readiness. The project demonstrated the importance of data quality and preprocessing before performing analytical operations.

This project also improved practical understanding of real-world data cleaning techniques and documentation standards used in Data Analytics workflows.